

PETROLEUM SYSTEM & GEOMECHANICAL RISK PROFILE

Geomechanical evaluation across Upper Assam fields establishes three primary drilling hazard windows: (1) Barail Group (1,800m–2,500m) containing 35–45% Smectite clay causing severe pipe sticking; (2) Kopili Formation (2,500m–3,200m) exhibiting pore pressure ramp up to 1.36 g/cc EMW requiring heavy mud weights; and (3) Sylhet Limestone (3,000m–3,500m) containing karstified vuggy voids leading to total mud loss.